ASSIGNMENT 1

1. Free Software Foundation/GNU is a nonprofit with a worldwide mission to promote computer user freedom which was founded in 1985 by Richard Stallman.It develops and promotes software and gives user freedom to run copy, study, modify and share.
   It is related to Linux in that Linux is the kernel which manages the computer's hardware and resources while GNU provides many of the tools e.g. bash needed to make a complete operating system.
2. A shell is a command line that takes commands and translates them for the operating system.
   It works with kernel in that it makes restricted requests called system calls to the kernel, preventing direct and unsafe user access to the hardware. Once kernel processes the request securely the shell receives the result and displays it, so it basically protects the hardware.
3. Linux is written in **C** language alongside with a small bit of assembly language. It makes Linux popular because it:

   - Is portable; compiles with almost any CPU architecture.
   - Has high performance; it translates directly into efficient machine code without performance penalties of virtual machine.
   - has large developer pool; many programmers know the language therefore easy to contribute.
4. cd/bin/cat will not work because bin/cat is a file not a directory while cd only changes directories.
5. /dev is a device node in which all devices, a hard drive or a printer, are seen as files and are accessed from - got this from the notes posted (File and Directories in UNIX.pdf pg9) /dev/null is a special device file that discards any data written to it, and is often called the null device or a black hole, for example, echo "Hello" > /dev/null sends the output nowhere, so it is not displayed – got this from:
   Linux Kernel Documentation, *Linux Allocated Devices*:
   https\://www\.kernel.org/doc/html/latest/admin-guide/devices.html
   Linux Manual Pages, *null(4)*: https\://man7.org/linux/man-pages/man4/null.4.html